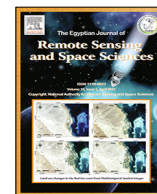


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Research Paper

Study of selected influential criteria on groundwater potential storage using geospatial technology and multi-criteria decision analysis (MCDA) approach: A case study [☆]

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ABSTRACT

Analytical hierarchical process (AHP) is nowadays a very popular tool in multi-criteria decision analysis (MCDA) in the groundwater suitability zone assessment. The present study involves the selection of six major influencing criteria, i.e., land use and land cover, drainage density, normalized difference vegetation index, lineament density, groundwater fluctuation, and the slope. Using remotely sensed data in a GIS environment, various thematic layers are generated. Suitable weights are assigned to these criteria using Saaty's scale as well as reclassification and ranking scores are also done to sub-criteria. Accordingly, consistency checks are also done to the weights applied to all the criteria. The weighted overlay technique is adopted under ArcGIS environment. Generated groundwater potential zone (GWPZ) map, illustrates five classified zones (1–5) of groundwater potentials, where 1 is representative of very poor prospects zone (10.9%), and 5 is indicative of excellent storage zones (37.8%).

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1. Introduction

Groundwater is considered as one of the most importantly concerning natural resources and its availability varies from place to another. Occurrences of groundwater storage below the earth's surface depend upon multiple factors. With the improvement of spatial accuracy in geospatial technologies, groundwater resources studies and research has gain appreciable momentum. Mapping of the groundwater requires the knowledge of hydro-physical characteristics of the area and is very complex (Sandoval and Tiburan, 2019; Zhang and Chui, 2019; Pourghasemi et al., 2020). Integrated and result-oriented nature of remote sensing and GIS have surely enhanced the capacity of the researcher (Abrams et al., 2018; Agarwal and Garg, 2016; Altafi Dadgar et al., 2020). Groundwater suitability and recharge zone delineation (Bonetto et al., 2020), vulnerability and risk zone assessment (Abu-Bakr, 2020; Butt et al., 2015), potential zone delineation and mapping (Mukherjee and Singh, 2020), groundwater contamination and quality assessment (Manikandan et al., 2020), groundwater monitoring

(Jarraya-Horriche et al., 2020), groundwater management (Sayed et al., 2019), and artificial recharge site selection etc. are the important aspects to be dealt with groundwater resources (Daesslé et al., 2020; Sallwey et al., 2019; Nithya et al., 2019).

In the recent trend, AHP become a very effective tool in solving the various environment-related issues, decision making, and prediction of the phenomenon. A broader range of applicability of the MCDA approach lies in the concept that it depends on the analyst defined set of parameters according to the environment (Cegan et al., 2017; Collins, 2007; Shadmehri Toosi et al., 2020). MCDA approach is also problem solving oriented to the groundwater phenomenon. Saaty is credited to develop AHP techniques (Saaty, 1987; Saaty and Varga, 2001), which involves the comparative weight assignment for selective influential parameters on the scale of 1–9 from equal importance to extreme importance respectively. The characteristics of AHP in MCDA also becomes inevitable since it is being quite capable of dealing of multiple factors of different units along with their pairwise comparisons at a time according to the weights allotted as per influences on the particular decision making (Leal, 2020; Kittipongvises et al., 2020; Ishizaka and Labib, 2011; Ananda and Herath, 2009).

Mirzapur district is heavily dependent on groundwater resources for its various purposes. Groundwater scarcity is well known and is considered the major problem of the district

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Abbreviations

AHP	Analytical hierarchical process	LU/LC	Land use and land cover
b.g.l	Below ground level	LD	Lineament density
CGWB	Central Ground Water Board	L_d, L_i	Lineament density, total length of lineaments (L_i) in unit area (A)
CI	Consistency index	MCDA	Multi-criteria decision analysis
CR	Consistency ratio	NDVI	Normalized difference vegetation index
DD	Drainage density	NRSC	National Remote Sensing Centre
D_d, D_i, A	Drainage density, total length of streams (D_i) in unit area (A)	NIR	Near- infrared radiation
EMR	Electromagnetic radiation	RI	Random index
GWPZ	Groundwater potential zone	SL	Slope
GWF	Groundwater level fluctuation	$\lambda_{\max}$	Eigenvalue

(<https://www.indiawaterportal.org>). The present study involves the delineation and mapping of groundwater potential in the sub-districts of Mirzapur district. The setting of preferences of topological and various hydrogeomorphological groundwater influential parameters in the study area is an important part of the present work through AHP in the MCDA approach. AHP technique is applied in the selected six major influential factors, namely, land use and land cover (LU/LC), drainage density (DD), normalized difference vegetation index (NDVI), lineament density (LD), groundwater level fluctuation (GWF), and slope (SL). A pairwise comparison matrix is prepared. Suitable weights are assigned to all the layers, and consistency ratio is computed, which is equals to 0.07. Reclassification, suitable rank scores, and weight assignment of sub-criteria and consistency check for all layers are respectively estimated. In the ArcGIS environment, all layers are added, and the weighted overlay tool is applied. The resultant overlay processed map consists of a delineated groundwater potential storage map. Class 1 is indicating low potential zone, whereas class 5 is expressed as very good potential zone in the region.

The GWPZ is also correlated with the proper marking of wells from the toposheet and the information collected through official

sites of the Bhuvan portal related to the groundwater prospect map. The part of Vindhyan upland exhibiting very poor zones. These regions are highly rugged and dissected. Very limited prospect is found near the valley regions of the upland part. Middle alluvial plain embraces the majority of marked well locations and during the analysis, these regions have exhibited good to very good prospect zones.

2. Materials and methods

2.1. Brief of the study area

2.1.1. Location

The longitudinal and latitudinal extent of the region is $82^{\circ}44'$ E to $82^{\circ}98'$ E and $24^{\circ}77'$ N to $25^{\circ}05'$ N respectively (Fig. 1).

2.1.2. Geomorphology

The region is approximately bordered by the highly rugged hilly Vindhyan upland in both the northern and the southern parts. Alluvial tract is running almost in the middle part of the region. A patch of the alluvial region also exists in the extreme northern

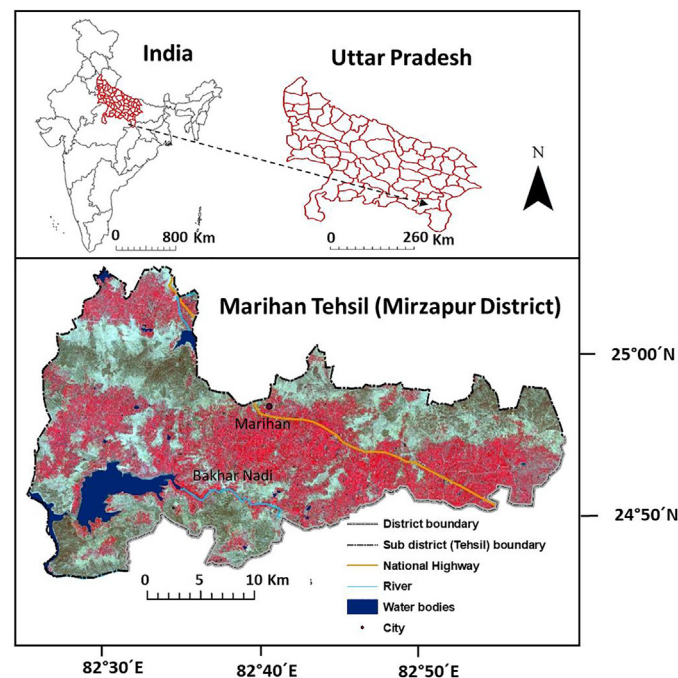


Fig. 1. Location of Marihan tehsil (Sub district of Mirzapur district, U.P.).

part. These two make broad physical division for the region. From geological point of view, alluvial tract belongs to Quaternary age with sand, silt, and clay composition. Upper Vindhyan plateau group composed mainly of quartzite, sandstone, and shale. Numerous characteristic geomorphic features are embedded with the region. It includes lineaments, valleys, waterfall, pediments, buried pediments, and residual hills, etc.

2.1.3. Climatic conditions

The region experiences three seasons predominantly, i.e., summer from mid-march to mid-June, winter from mid-October to February, and southwest monsoon season from mid-June to September. Summer is very hot and dry. Maximum rainfall occurs during the rainy winter months. The average annual rainfall is about 975 mm. The region has comparatively more rainfall in winter than the summer.

Table 1
Data source and their description.

Data type	Description/Resolution/Year	Source
Satellite	Resourcesat 2 (IRS P6, LISS-III)/ 23.5 m/ 2017	Bhuvan- NRSC
DEM	Cartosat 1/ 2.5 m	Bhuvan-NRSC
Topographical sheet	Official website	Survey of India portal
Pre and post monsoon groundwater level, Groundwater yield and prospect information	Official website	Ground Water Department, Uttar Pradesh, Central Ground Water Board (CGWB), Bhuvan-bhujal
Climatic Information	Official website	Indian Meteorological Department, Ground Water Department, Uttar Pradesh
Geology, geomorphology		District Gazetteer

2.2. Data source

Table 1 shows the list of collected data from the various sources and their detailed description.

2.3. Methodology

2.3.1. Thematic layer preparation

Toposheets of the study area are brought in the ArcGIS environment and georeferenced at WGS –84 datum. An area of interest is subsequently extracted. A subset from both the satellite image with standard FCC and DEM is also created. For the GWPZ study, land use and land cover, drainage density, NDVI, lineament density, groundwater fluctuation, and the slope are taken as influencing factors (Fig. 2).

2.3.2. Criteria weight assignment

AHP in multi-criteria decision analysis is a very popular method to give weight to different factors according to their relative importance. Saaty's (1987) scale of 1–9 of comparative rating is adopted for relative comparison (Table 2). Secondary sources of information like available literature, official sites of central and state Groundwater Board and relevant thematic maps from Bhuvan portal, topographical sheet study, and the satellite images interpretation are

Table 2
Saaty's (1987) scale for pairwise comparison.

Intensity of influence	Explanation
1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong importance
9	Extreme importance
2, 4, 6, and 8 are the intermediate values	

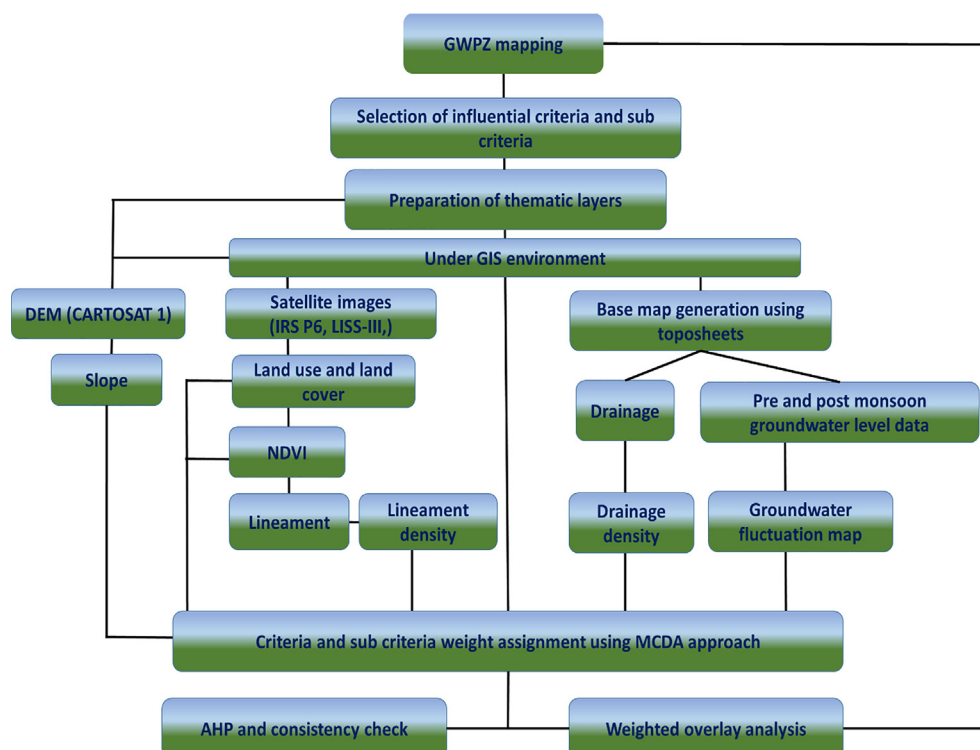


Fig. 2. Data sources and methodology.

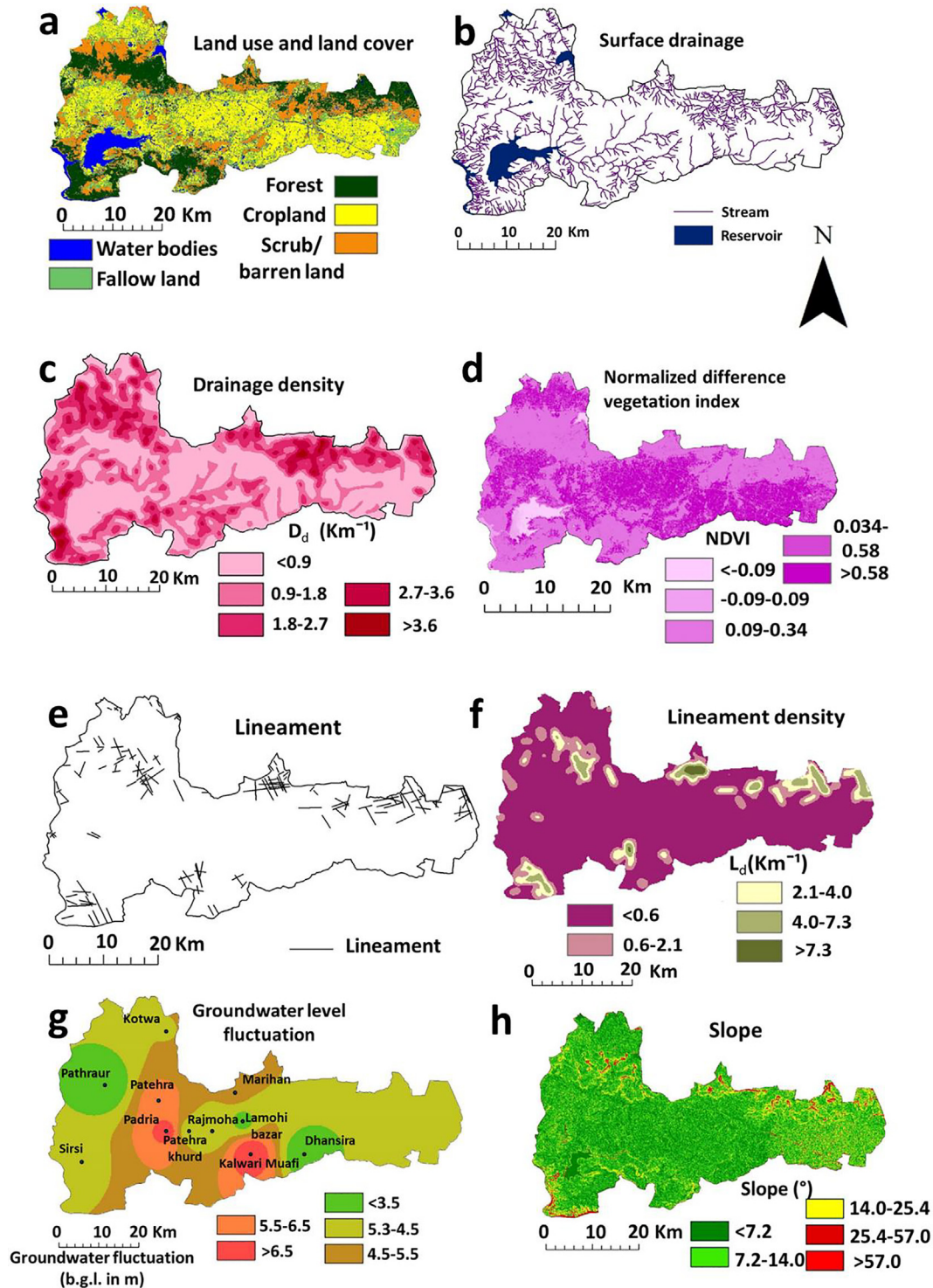


Fig. 3. (a): LU/LC, (b): Surface drainage, (c): DD, (d): NDVI, (e): lineament, (f): LD, (g): GWF, (h): Slope.

the basis of criteria's weight assignment. Selected all six basic parameters are compared on Saaty's scale and relative weights/scores are given accordingly. A pairwise comparison matrix is generated to compute further eigenvector, consistency ratio, and normalized weight. The consistency index and consistency ratio is computed by using the Eq (1) & (2) since according to Saaty's pairwise comparisons are consistent with the value < 0.10 (Ahmad et al., 2020; Gdoura et al., 2015; Murmu et al., 2019).

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (1)$$

$$CR = \frac{CI}{RI} \quad (2)$$

where, CI is the consistency index, $\lambda_{\max}$ is an eigenvalue, n is the number of influencing criteria, CR is the consistency ratio, and RI is the random index with value of 1.24.

Table 3
LU/LC classes: Marihan tehsil.

LU/LC classes	Area (%)
Scrub/barren land	21.13
Fallow land	19.49
Forest	20.26
Cropland	30.69
Water bodies	8.43

Table 4
DD classes: Marihan tehsil.

DD (km ⁻¹)	Description
0–0.9	Very low
0.9–1.8	Low
1.8–2.7	Moderate
2.7–3.6	High
3.6–4.5	Very high

Table 5
NDVI classes: Marihan tehsil.

NDVI value	Description
<-0.09	Very low
-0.09–0.09	Low
0.09–0.34	Moderate
0.34–0.58	High
>0.58	Very high

Table 6
LD classes: Marihan tehsil.

LD (km ⁻¹)	Description
0–0.6	Very low
0.6–2.1	Low
2.1–4.0	Moderate
4.0–7.3	High
7.3–12.2	Very high

Table 8
GWF classes: Marihan tehsil.

GWF (b.g.l in m)	Description
<3.5	Very low
3.5–4.5	Low
4.5–5.5	Moderate
5.5–6.5	High
>6.5	Very high

Table 9
Slope classes: Marihan tehsil.

Slope (°)	Description
<7.2	Level to gentle
7.2–14.0	Gentle
14.0–25.4	Moderate
25.4–57.0	Steep
>57.0	Very steep

Table 10
Pairwise comparison matrix and normalized weight of influencing parameters.

	LU/LC	DD	NDVI	LD	GWF	SL	Normalized weight
LU/LC	1	3	5	5	4	5	0.43
DD	1/3	1	2	3	5	5	0.23
NDVI	1/5	1/2	1	1	3	4	0.12
LD	1/5	1/3	1	1	2	5	0.11
GWF	1/4	1/5	1/3	1/2	1	3	0.07
SL	1/5	1/5	1/4	1/5	1/3	1	0.04

total number of criteria selected and the total number of classes in each criteria layer respectively.

3. Result and discussion

3.1. LU/LC

Using visual image interpretation, carefully training areas for the different land use categories are marked. The maximum likelihood classifier is run for the supervised classification. The resultant image is expressed into scrub/barren land, fallow land, forest, cropland, and water bodies (Fig. 3(a)). Forest is occupied almost on the hilly regions having 20.26% of the total area. An appreciable share of scrub/barren land also exists with 21.13% of the total area. The middle portion, extreme north, and buried pediment zones are the cultivated region of the area. Cropland has maximum area cover from rest of the class with 30.69% of the share. Fallow land

2.4. Groundwater potential zone

$$GWPZ = \sum_{i=1}^m \sum_{j=1}^n (M_i \times R_j) \quad (3)$$

where, M_i is the normalized weight of the i^{th} criteria layer, R_j is the normalized weight of the j^{th} class of criteria layer, m & n are the

Table 7
Pre monsoon and post monsoon groundwater level fluctuation for the year 2014 and 2017 (b.g.l. in m).

Station	Location	2014		2017		Average fluctuation	
		Pre monsoon	Post monsoon	Pre monsoon	Post monsoon		
Jamuee bazar	82°70'E 24°89'N	3.85	2.50	1.35	6.85	4.80	3.07
Kotwa pandey	82°60'E 25°01'N	5.90	1.30	4.60	6.35	5.10	4.85
Marihan bazar	82°69'E 24°93'N	7.18	2.80	4.38	9.53	6.50	5.44
Marihan block (patehra)	82°59'E 24°92'N	14.64	9.84	4.80	16.84	8.39	6.59
Padreeya	82°66'E 24°88'N	5.05	4.25	0.80	10.00	3.25	3.77
Pathehra palnagar	82°63'E 24°88'N	7.34	4.34	3.00	8.54	3.74	3.90
Patharaaur	82°52'E 24°94'N	4.55	3.45	1.10	4.60	2.57	1.56
Rahoja	82°60'E 24°88'N	8.55	4.11	4.44	15.90	3.85	12.05
Sirshi	82°49'E 24°84'N	9.30	4.75	4.55	dry	4.25	4.40
Kalwari Mufi	82°74'E 24°85'N	16.32	6.47	9.85	15.47	6.12	9.35
Dhansira	82°71'E 24°85'N	3.60	3.60	0.00	7.20	2.65	4.55

associated with the cultivated region and water bodies shares nearly 19.49% and 8.43%, respectively of the total area (Table 3).

3.2. Drainage density

From Fig. 3(b), it can be clearly interpreted that the hilly region has a high frequency of the drainage, and streams are closely spaced as well as density is also high. The alluvial area is showing less drainage density and is dendritic in shape. Depending upon the value, computed using the Eq. (4), the D_d map is categorized into very low, low, moderate, high, and very high drainage density (Fig. 3(c) and Table 4).

$$D_d = \sum_{i=1}^n \frac{D_i}{A} \quad (4)$$

where, D_d is the drainage density, D_i is the total length of streams in the unit area A .

3.3. Normalized difference vegetation index (NDVI)

NDVI is one of the important indicators that involves the study of the behaviour of the vegetation to the EMR. This numerical index is quite useful in monitoring the vegetation covers over the surface of the earth. NDVI value ranges from -1 to $+1$ and is calculated using the Eq. (5).

$$NDVI = (NIR - RED)/(NIR + RED) \quad (5)$$

The negative values are indicative of no vegetation cover like bare areas, water bodies and etc., whereas higher positive value articulates the high vegetation cover over the region.

Table 11
Individual rank and weight assignment for six major influencing parameters: Marihan tehsil.

Influencing parameters	Sub-categories	Consistency ratio	Rating	Normalized weight
LU/LC	Scrub/ barren land	0.05	1	0.06
	Fallow land		2	0.06
	Forest		3	0.19
	Cropland		4	0.22
	Water bodies		5	0.47
DD	<0.9	0.04	5	0.31
	0.9–1.8		4	0.31
	1.8–2.7		3	0.24
	2.7–3.6		2	0.09
	>3.6		1	0.05
NDVI	<-0.09	0.05	1	0.07
	-0.09–0.09		2	0.07
	0.09–0.34		3	0.13
	0.34–0.58		4	0.34
	>0.58		5	0.39
LD	<0.6	0.01	1	0.07
	0.6–2.1		2	0.08
	2.1–4.0		3	0.11
	4.0–7.3		4	0.34
	>7.3		5	0.40
GWF	<3.5	0.07	5	0.37
	3.5–4.5		4	0.29
	4.5–5.5		3	0.19
	5.5–6.5		2	0.09
	>6.5		1	0.06
SL	<7.2	0.07	5	0.40
	7.2–14.0		4	0.31
	14.0–25.4		3	0.16
	25.4–57.0		2	0.08
	>57.0		1	0.05

In the study region, dissected plateau over the region showing the negative and low value of NDVI, whereas alluvial regions have high value with >0.58 (Fig. 3(d) and Table 5).

3.4. Lineament density

Lineament is considered as a very important geomorphic feature and tool for the study of groundwater resource prediction in hard rock regions below the earth's surface. The presence of these linear lineaments and fractures on hard rock terrain are the indicative of the presence of good groundwater storage. High L_d has higher chances of groundwater potential. The feature is carefully marked on the imagery. The varying density value is grouped into five categories from very low to very high density using the Eq. (6) (Fig. 3(e) and (f) and Table 6).

$$L_d = \sum_{i=1}^n \frac{L_i}{A} \quad (6)$$

where, L_d is the lineament density, L_i is the total length of lineaments in the unit area A .

3.5. Groundwater level fluctuation

Eleven hydrograph stations are taken to study the fluctuation rates in the present region. From the average values of the data of the two years 2014 and 2017 (Table 7), Patharaur with an average value of 1.6 m b.g.l showing less fluctuation compared to Kalwari Mufi with the high average value of 9.6 m b.g.l. The region is displayed into five categories of fluctuation level. Value < 3.5 m b.g.l. to value > 6.5 m b.g.l is scaled with very less to very high fluctuation (Fig. 3(g) and Table 8).

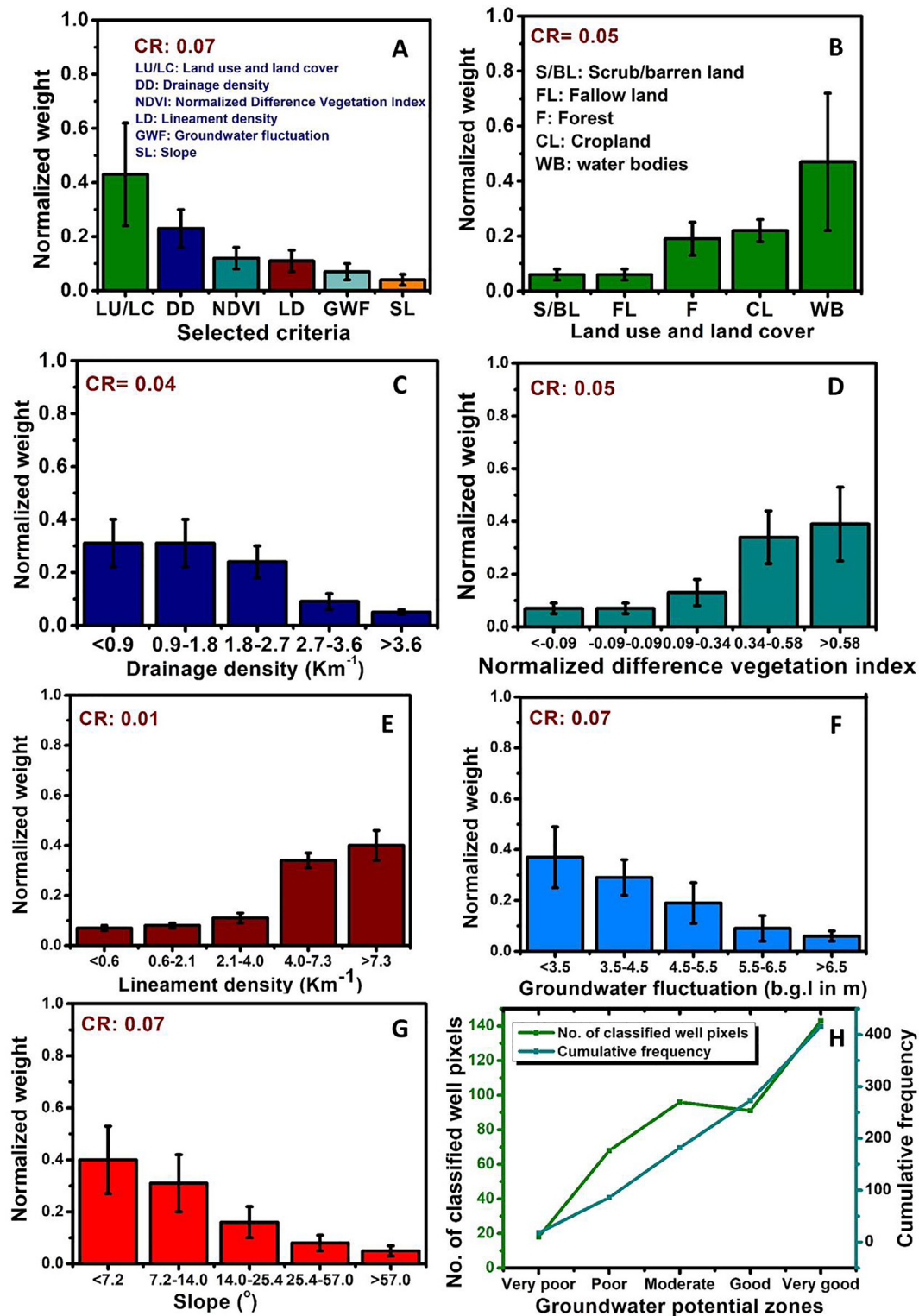


Fig. 4. Normalized weight and CR (A): Criteria whole (0.07), (B): LU/LC (0.05), (C): DD (0.04), (D): NDVI (0.05), (E): LD (0.01), (F): GWF (0.07), (G): Slope (0.07), (H): Graph showing well pixels to the respective GWPZ.

3.6. Slope

Slope is also considered as one of the most influential determinants in groundwater potential storage. The steeper slope leads to higher chances of runoff, consequently fewer chances for the infiltration. Hence the high potential for groundwater storage is suppressed in hilly regions. In the study, the slope is kept into five classes with levels from gentle to the very steep slope (Fig. 3(h) and Table 9).

3.7. Pairwise comparison matrix and consistency check

All the six major influential parameters are scaled and a comparison is made pairwise to each factor. Table 10 shows the respective weights of each factor according to their importance of influence.

The comparison is consistent if CR value ≤ 0.1 . The eigenvector value for the present comparison matrix (Table 10) is estimated

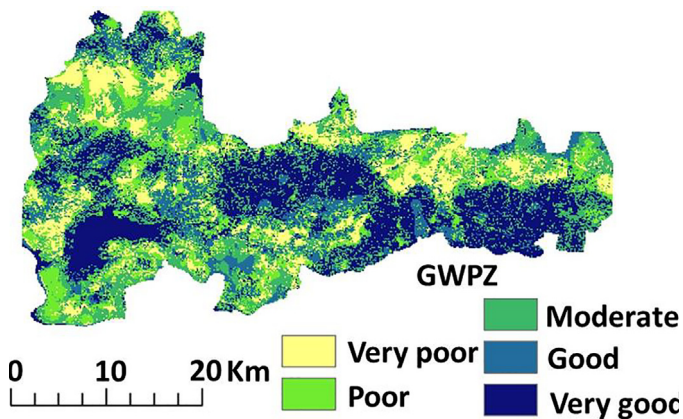


Fig. 5. GWPZ classes: Marihan tehsil.

equals 6.4. Applying Eqs. (1) and (2), CR value in the present case in 0.07, which is <0.10 . Hence the comparisons are consistent with the weights applied. 43, 23, 12, 11, 7, and 4% are weights are given to LU/LC, DD, NDVI, LD, GWF, and SL respectively. All the thematic influencing layers are again reclassified. The sub-categories are assigned individual weights according to their intensity of influence. Individual rating, normalized weights, and consistency check for each sub-criteria of all major factors are estimated and shown in Table 11 and Fig. 4(A), (B), (C), (D), (E), (F), and (G).

4. GWPZ

With the application of weighted overlay technique tool to all the thematic layers with their suitable rankings under the ArcGIS environment, the GWPZ map is generated (Fig. 5). A generated map with a 1 to 5 scale consists of 1 for poor prospect zone and

Table 12
GWPZ, yield and prospect zone description.

GWPZ	Area (%)	Yield zone range (Source: Bhuvan-bhujal, in LPM)	Description	Characteristics
Very poor	10.9	20–30	No yield, >80 m deep well	Dissected hilly region, highly steep, high drainage density, low lineament density
Poor	18.0	20–30	>80 m and 30–80 m deep well	Dissected hilly region, again scattered north and south and middle of the region
Moderate	23.3	20–30	>80 m and 30–80 m deep well	Associated with the very poor and poor prospect zones
Good	10.0	30–50, 50–100	>80 m deep well	Found in the patches near the reservoir and associated with good very good zones for the prospects. Buried zones are marked features
Very good	37.8	50–100, 100–200, 200–400	>80 m deep well, 30–80 m deep well, <30 m deep well	Associated with the alluvial plain, water bodies. Patches lie almost in entire middle and south western portion of the region.

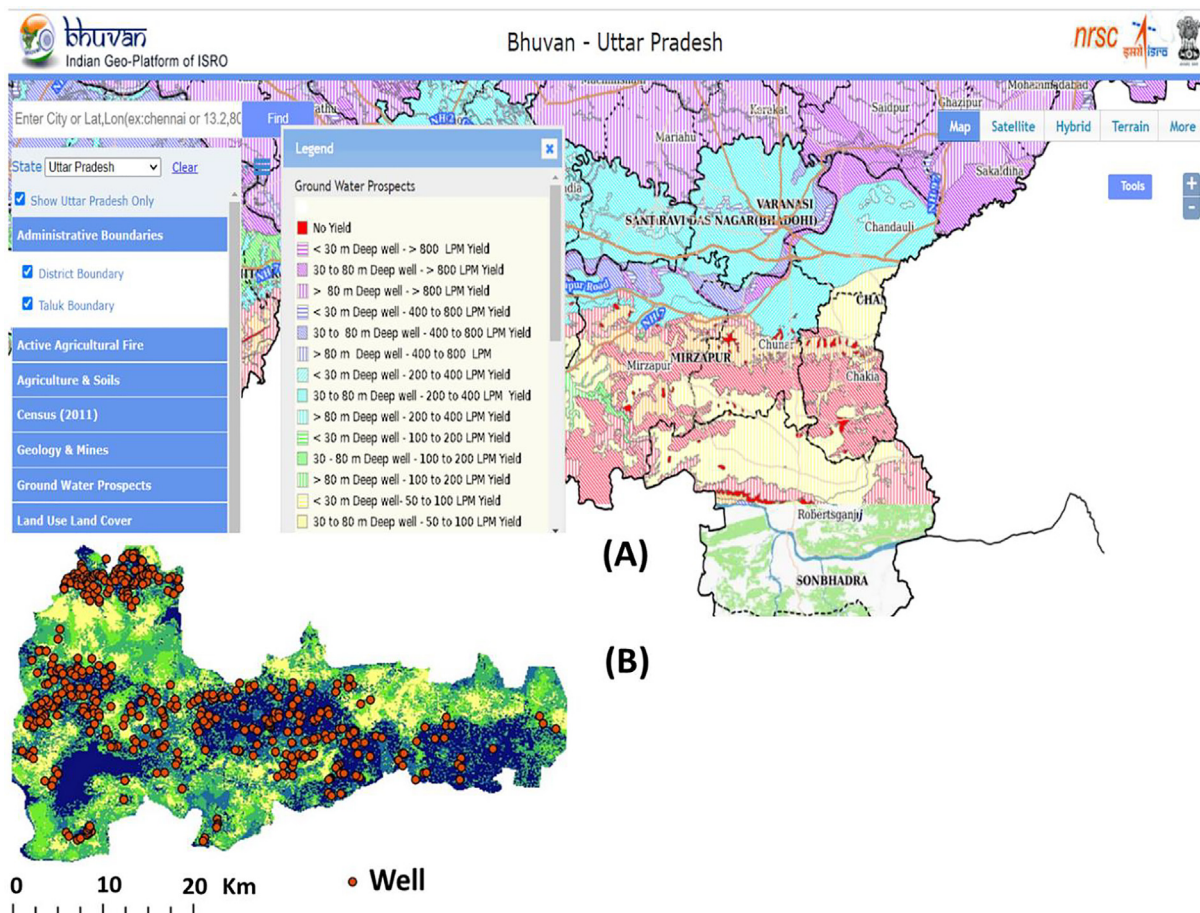


Fig. 6. GWPZ study from wells location in respective zones and Bhuvan-bhujal groundwater prospect for the Marihan tehsil.

value 5 for very good potential zones for the groundwater storage. A very poor to poor prospect is found in the areas of the hilly dissected plateau region. Good to very good zones are the areas of buried zones and alluvial areas.

In Table 12, GWPZ is also studied taking yield (LPM) of the respective zones from the Bhuvan bhujal portal, which indicates a similar depiction. The presence of wells is considered as an important indicator of groundwater storage potential of the region. Total of 416 wells are carefully marked on the region from the toposheet of the study area. The location of well points is overlaid on the predicted resultant GWPZ map to analyze the groundwater prospect zones. From the well pixels, it is very clear that the majority of the points are coming under the classified categories of moderate to very good potential zones (Fig. 4(H)).

The yield of very poor to poor prospect ranges only 20–30 LPM. Wells are noted even >80 m deep. In the moderate zones depth of well ranges from 30 to 80 m and > 80 m. Yield from the reference is noted under 20–30 LPM. In the good to very good zones yield ranges are found 50- to even 400 LPM. The range of the well depth is 30–80 m and also < 30 m (Table 12 and Fig. 6).

5. Conclusion

The very popular method of MCDA analysis is quite useful in the prediction of GWPZ delineation and mapping. Six parameters were taken, namely, LU/LC, DD, NDVI, LD, GWF, and slope. AHP tool is applied and the consistency check is done. It is 0.07 for major influencing criteria. Weighted overlay technique involves reclassification and suitable weight assignment of sub-criteria of each major parameter. Consistency ratio is also computed for each sub-criteria, which are 0.05, 0.04, 0.05, 0.01, 0.07, and 0.07 for LU/LC, DD, NDVI, LD, GWF, and slope, respectively. The resultant GWPZ map consists of five zones classified from 1 to 5 representing low to high prospects for the groundwater potential storage. Wells are marked on this map to assess the classified suitability of each zone. It is found that the maximum concentration of wells is lying in the middle alluvial plain ranging from moderate to very good zones. A reference map from the official site from Bhuvan bhujal is also taken to assess the result in terms of the yield and the prospect storage. Dissected plateau region bordering north and south and scattered in the middle also, are very poor to poor yield zones and hence have a poor capacity for groundwater storage.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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